Learned-model datasets — inverse dynamics (ID) · forward dynamics (FD) · reduced-order model (ROM)

Closed-loop campaign · realistic operation (R1, R3)

Latin-hypercube parameter sampling (12-D)

Fixed integral-backstepping controller

Waypoint trajectory sampler (12–20 poses)

1000-run curriculum, ode23t integration

Per-run logging: state, command, ξ

Randomization

Align & Join

Open-loop campaign · excitation (R2)

64 s open-loop excitation schedule

Four plant draws (nom / hyd / mech / full)

Stroke-coverage acceptance gate

Rejected export — duplicated-state audit fail

Audit & Validation

Hydraulic subsystem

Multibody subsystem

Simscape model interface

CAD / URDF geometry

ode23t stiff solver

Nominal parameter set